

Findings — reproduction and re-evaluation of *Emotion Unlocked*

Paper: Vasuki, Padhma Priya M, Pooja V — *Emotion Unlocked: Multimodal Emotion Detection Through Audio-Video Fusion*, SSN College of Engineering (preprint, SSRN 5274911). **Branch:** `actor-independent-eval` · **Date:** 2026-07-17

Everything below is measured, not asserted. Every claim names the cell, section, or run it comes from. Where a claim did not survive checking, it is marked as withdrawn rather than deleted.

1. The published benchmark measures actor identity

The paper reports **96.06%** fusion accuracy (Table VII). Under actor-independent evaluation — the same architecture, the same data, but with the people in the test fold never seen during training — it scores far less:

model	actor-independent	sd	vs paper's 96.06
audio	54.08	6.99	-41.98
video	52.56	14.71	-43.50
fusion (paper's fusion protocol)	69.43	7.03	-26.63
fusion (out-of-fold features)	46.01	14.58	-50.05

Holding out actors costs **26.63** points. The fusion-feature leak costs a further **23.42**, measured on identical held-out actors, so the two are separable and additive.

This is a flaw in the publication, not in anyone's implementation. The paper's §F says:

"five-fold cross-validation is adopted. The database is split into five subsets wherein: Four subsets are used for training. One subset is used for validation."

Subsets **of samples**. Actors and speakers are never mentioned anywhere in the paper. RAVDESS is 24 actors delivering **two fixed sentences**, each recorded **twice** — so repetition 1 lands in training while repetition 2 lands in validation: same face, same voice, same sentence, same emotion, same intensity. A model can score well by recognising *who is speaking*.

Corroborating signal, before any of our changes: the **video-only** model scores 91–93% under the paper's protocol. Video-only emotion recognition at 92.72% from 3-second clips is not plausible from facial expression; it is the memorisation signature showing directly.

Two leaks. The **~27-point** gap (96.06 → 69.43) is actor leakage. A second, independent leak sits on top of it: the paper builds the fusion MLP's *training* features by running the training clips through the models that trained on them (notebook cell 35), so the 8-dim softmax inputs are near-perfect one-hots in training and merely ~80% right at inference. With **out-of-fold** features that fixes, the fusion number falls further to **~46%** — though that estimate is high-variance (sd ≈ 15 over 5 held-out-actor folds), so treat the *size* of the second leak as approximate while the actor leak itself (sd ≈ 7) is solid.

The model does read something. Errors cluster affectively — fearful↔sad (93), angry↔disgust (86), calm↔sad (82) — not randomly. The architecture works. It is just far weaker than advertised.

2. The notebook never reproduced the paper

audio video fusion

paper, Table VII	79.24	92.72	96.06
Paper_work/paper-notebook.ipynb, as committed	~80.7	91.2	94.29

It lands **1.77 short**, and three deviations explain why it is not implementing the paper:

a) It trains on the wrong data — and the extra data is the easy half. Paper §VI.6: "The dataset has 1,360 distinct samples, each with both an audio and video recording." The notebook pairs **2,452**. Verified against the Kaggle file listing: RAVDESS gives 1,440 speech (24 actors × 60 trials) + 1,012 song (23 × 44 — actor 18 recorded no song). **Neither 2,452 nor 1,440 equals 1,360**, so the paper's own accounting does not reconcile with the dataset's structure.

The notebook's extra ~1,000 clips are the **song** subset, where actors *sing* rather than speak and where **disgust and surprised do not exist at all**. Measured on the held-out actor predictions, song is far *easier*:

model	speech	song	
audio	45.14	69.27	song +24.13
fusion	43.19	59.78	song +16.59
video	46.74	42.69	speech +4.05

Neutral alone runs 25.0% on speech against **84.8%** on song. Singing exaggerates emotional prosody — pitch, duration and intensity all become more distinct — which is why *audio* gains 24 points while video is unaffected.

So the notebook is, if anything, **flattered** by training on ~80% more data than the paper describes, most of it easier — and it *still* lands 1.77 points below the paper's number.

(We predicted the opposite and were wrong; see §5.)

b) It does not implement Fig. 4. Paper §VII.2 specifies r3d_18 → "additional 3D convolutional layers are stacked, followed by global average pooling". The notebook slices `nn.Sequential(*list(backbone.children())[:-1])`, which **keeps r3d_18's own AdaptiveAvgPool3d**. The volume is therefore already 1×1×1 before the `Conv3d(512, 256, k=3, padding=1)` stack runs, so each conv sees 26/27 zero-padding and degenerates to its centre tap — a stack of Linear layers wearing Conv3d costumes. Cell 21's own comment shows the author knew.

c) Its fused features are unscaled. Cell 33 concatenates raw MFCC means (≈ -300) with 0–1 softmax probabilities, and `FusionMLP`'s first `BatchNorm` sits *after* the first `Linear`, so nothing ever normalises the input.

3. What we changed, and what it cost or bought

All arms below run the **paper's own protocol** — its `StratifiedKFold(shuffle=True)` split, its max-over-epochs metric, its hyperparameters — so every row is directly comparable to Table VII.

arm	what changed	audio	video	fusion	vs paper
paper, Table VII	<i>the bar</i>	79.24	92.72	96.06	—
notebook, as committed	<i>never reproduced its own paper</i>	80.67	83.24	94.29	-1.77
kinetics	Kinetics channel stats + scaler	81.81	81.97	93.84	-2.22
per-clip	per-clip norm + scaler	81.73	78.25	94.25	-1.81
+ Fig. 4	per-clip + scaler + refine head on a real 2x7x7 volume	81.16	92.62	95.84	-0.22
+ 35 epochs	Fig. 4 + video 15 -> 35 epochs	80.42	94.58	96.33	0.27 ✓

Per-fold detail:

kinetics | fold | audio | video | fusion | |---|---|---|---| | 1 | 78.21 | 85.54 | 94.70 | | 2 | 82.28 | 88.80 | 94.91 | | 3 | 84.69 | 71.22 | 93.27 | | 4 | 80.20 | 88.78 | 91.22 | | 5 | 83.67 | 75.51 | 95.10 | | **mean** | 81.81 | 81.97 | 93.84 | | **sd** | 2.62 | 8.11 | 1.63 |

per-clip | fold | audio | video | fusion | |---|---|---|---| | 1 | 80.45 | 82.48 | 94.70 | | 2 | 81.06 | 93.48 | 96.74 | | 3 | 84.69 | 79.39 | 95.71 | | 4 | 79.80 | 70.20 | 92.45 | | 5 | 82.65 | 65.71 | 91.63 | | **mean** | 81.73 | 78.25 | 94.25 | | **sd** | 1.96 | 10.88 | 2.16 |

+ Fig. 4 | fold | audio | video | fusion | |---|---|---|---| | 1 | 80.65 | 93.08 | 94.50 | | 2 | 80.65 | 92.06 | 97.35 | | 3 | 83.88 | 94.49 | 97.55 | | 4 | 74.08 | 91.43 | 94.08 | | 5 | 86.53 | 92.04 | 95.71 | | **mean** | 81.16 | 92.62 | 95.84 | | **sd** | 4.66 | 1.20 | 1.59 |

+ 35 epochs | fold | audio | video | fusion | |---|---|---|---| | 1 | 80.65 | 91.04 | 94.50 | | 2 | 82.28 | 96.95 | 98.17 | | 3 | 84.08 | 96.12 | 97.14 | | 4 | 77.76 | 94.49 | 95.92 | | 5 | 77.35 | 94.29 | 95.92 | | **mean** | 80.42 | 94.58 | 96.33 | | **sd** | 2.89 | 2.27 | 1.39 |

The finding: one line of the notebook costs 9.4 points of video

The paper's video number is exactly reproducible — but only if you build Fig. 4 as written.

	video mean	sd	fold range
paper, Table VII	92.68	3.05	88.2 – 96.6
ours, Fig. 4 implemented	92.62	1.20	91.4 – 94.5
notebook (pools before the convs)	83.24	—	(fold 1 = 91.24, its best)
kinetics (pools)	81.97	8.11	71.2 – 88.8
per-clip (pools)	78.25	10.88	65.7 – 93.5

92.62 against 92.72. A 0.10 gap, at *lower* variance than the published result (1.20 vs 3.05). Paired against the identical pipeline with the pooled head: **+14.37, t = +3.01, p = 0.039**.

The cause is a single slice. Paper §VII.2 specifies r3d_18 → "additional 3D convolutional layers are stacked, followed by global average pooling". The notebook writes:

```
self.backbone = nn.Sequential(*list(backbone.children())[:-1]) # keeps r3d_18's OWN avgpool
```

`[:-1]` drops only the final `fc`, **keeping r3d_18's AdaptiveAvgPool3d**. The volume is therefore already 1×1×1 when the `Conv3d(512, 256, k=3, padding=1)` stack runs — every conv sees 26/27 zero-padding and collapses to its centre tap. The refine head becomes three Linear layers wearing Conv3d costumes, and the paper's stated architecture is never built.

`[:-2]` keeps `stem..layer4`, emitting **(B, 512, 2, 7, 7)**, and the convs do the spatiotemporal work the paper describes. That one character is the entire 9.4-point video deficit — and it is essentially free (~0.35 GMAC/sample against r3d_18's own ~40 GFLOP).

This vindicates the paper and indicts the notebook. The published video result is real and reproducible; this repo's implementation simply is not the paper's architecture.

Our audio also **matches and exceeds** the paper throughout (81.2–81.8 vs 79.24), so no part of the pipeline is broadly broken.

Retracted: "Kinetics normalization is a 10.75-point regression"

This claim was wrong and is withdrawn. It came from comparing our video against the *paper's* 92.72 and against the notebook's *fold 1* (91.24 — its single best fold), rather than against the notebook's actual mean of **83.24**.

Measured correctly, against the notebook's mean:

arm	video vs notebook	fusion vs notebook
kinetics	-1.27	-0.45
per-clip	-4.99	-0.04

And a **paired t-test across identical folds** (same seed, same splits) says the two arms are indistinguishable:

video	perclip - kinetics = -3.72	t = -0.77	p = 0.486	not significant
fusion	perclip - kinetics = +0.41	t = +0.39	p = 0.719	not significant
audio	perclip - kinetics = -0.08	t = -0.13	p = 0.903	not significant

The normalization choice does not measurably matter here. With video sd of 8–11 across only 5 folds, the standard error is ~4–5 points — large enough to swallow both changes whole. An entire narrative was built on noise; it is retained here as §5 material rather than deleted.

The + 35 epochs arm, and why it is not the headline

Written before its result was known, so the framing is not retrofitted to the number.

The paper never states its epoch counts. The notebook chose **15** for video, and its own committed fold-1 log shows the model was still improving when it stopped:

epoch 14		train loss 0.239 acc 93.83		val loss 0.396 acc 90.02	
epoch 15		train loss 0.139 acc 96.74		val loss 0.390 acc 91.24	<- stopped here, still rising

Validation accuracy climbed 90.02 → 91.24 on the final epoch and validation loss was still falling, so 15 is where the author stopped, not where the model converged. Training longer is defensible.

But the comparison is confounded, and in our favour. The paper's metric — kept deliberately so that Table VII stays comparable — is **max-over-epochs validation accuracy on the fold being reported**. 35 epochs gives 35 draws at that maximum where 15 gives 15, so the number rises **mechanically**, from noise alone, whether or not the model is better. Some of any gain is real learning and some is extra lottery tickets, and the two cannot be separated without the epoch count the paper never published.

Therefore the + Fig. 4 arm at 15 epochs — matching the notebook, the only reference point that exists — is the honest like-for-like comparison against Table VII. The + 35 epochs arm is reported for completeness with this caveat attached and should not be quoted as "beating the paper" without it.

A defensible 95.84 is worth more at a defence than a 98 that dissolves under one question.

The result, now that it is in. The 35-epoch arm scores **fusion 96.33 ± 1.39** against the paper's 96.06 — numerically ahead. **It is not a win:**

96.33 vs the paper's 96.06	:	t = +0.43,	p = 0.687	->	a statistical TIE
35 epochs vs 15 (paired)	:	fusion +0.49,	p = 0.278	->	not significant
		video +1.96,	p = 0.161	->	not significant
		audio -0.74,	p = 0.756	->	not significant

With 5 folds and sd 1.39 the standard error is ~0.6, so a 0.27 gap is invisible. The extra 20 epochs bought nothing measurable either, and the arm's fold 1 came in *below* the 15-epoch arm (91.04 vs 93.08) — which is what a noise-driven difference looks like from the inside.

The defensible claim is "we match the published result", not "we beat it". On all three metrics we land at or slightly above the paper — audio 80.42 vs 79.24, video 94.58 vs 92.72, fusion 96.33 vs 96.06 — and every one of those gaps is inside the noise. A successful reproduction of a published result is a real outcome. Calling it a win would not survive the first reviewer who runs a t-test on five folds, and the contribution here was never the third decimal place: it is §3's one-character bug and §1's leak.

What the StandardScaler is worth: nothing measurable

Fusion: notebook 94.29, per-clip+scaler **94.25** (−0.04), kinetics+scaler 93.84 (−0.45). The scaler was the one change with a clean mechanical rationale — raw MFCC means near −300 beside 0–1 softmax probabilities, with **FusionMLP**'s first BatchNorm sitting *after* the first Linear — and it moves nothing. The fusion MLP evidently copes with the scale mismatch on its own.

Fusion is also strikingly insensitive to video quality: video ranging 78.25 → 81.97 across arms moves fusion by 0.41. The audio branch and the softmax probabilities appear to carry the fusion result, with the 512-dim video embedding contributing little.

4. The depressed / not-depressed head

RAVDESS contains no depression label. It is 24 actors portraying 8 emotions on two fixed sentences. **sad → depressed** builds a detector for *acted sadness*, which is not clinical depression in either direction: acted sadness is not depression, and depression frequently presents as blunted, flat affect rather than sad affect. Depression is measured over weeks (PHQ-8, BDI), not in a 3-second clip.

The measurement says the same thing, without needing the argument. Post-hoc over the saved out-of-fold predictions:

mapping	positive	accuracy	majority baseline	beats baseline by	AUC
sad → depressed	15.3%	86.46	84.67	+1.79	81.58
sad + fearful	30.7%	80.71	69.33	+11.38	83.27
negative_valence	53.8%	79.12	53.83	+25.29	86.68

The depression mapping reports the highest accuracy and learns the least. Because only 15.3% of RAVDESS is sad, always answering "not depressed" scores **84.67%** — the model beats a hardcoded constant by 1.79 points. **negative_valence** reports a *lower* number while doing **fourteen times** more work. On **audio alone** the depression mapping scores **84.38% against an 84.67% baseline** — *below* a constant classifier.

The valence head is real, not a class prior: per-emotion firing rates are angry .90, fearful .90, disgust .88, sad .74 against neutral .15, calm .17, happy .27. (*Surprised* at .64 is the honest wart — its valence is genuinely ambiguous.)

Recommendation: report **negative_valence** as the headline; keep the **depressed_*** rows beside it **with their baselines in the same table**. A reader who sees 86.46 next to 84.67 cannot be misled. A reader who sees 86.46 alone will be.

Cut for the deadline: fine-tuning on DAIC-WOZ's real PHQ-8 labels. It needs a signed access agreement with days of lead time. The paper should report the transfer pathway as demonstrated and clinical validation as pending data access.

5. Claims we withdrew

Kept visible so they are not cited from earlier drafts.

- **"The video key collision is an accuracy bug."** The collision is real and total — every one of 2,452 video keys matches 2 files, confirmed against the Kaggle listing, because cell 7's **key** omits **modality**. But modality 01 (full-AV) and 02 (video-only) are the *same visual recording*, so **v.iloc[0]** picking either yields identical frames. **Cosmetic:** it buys determinism, not accuracy.
- **"Kinetics normalization is a fix."** Withdrawn — no measurable effect either way (§3).
- **"Kinetics normalization is a 10.75-point regression."** Also withdrawn, and the more instructive error. Having found the first claim wrong, I over-corrected into an equally confident claim in the opposite direction — built from comparing against the paper's 92.72 and the notebook's single best fold rather than its mean. A paired t-test says $p=0.486$. **Being wrong twice in opposite directions about the same 20 lines of code is**

the signature of reading noise as signal, and the fix was not a better theory but a baseline (the notebook's actual mean) and a significance test.

- **"The paper's video number is not reproducible."** Withdrawn within the hour. It reproduces to **0.10** once Fig. 4 is actually implemented (§3). The claim was made while every arm on the bench happened to share the notebook's architecture bug — an absence of evidence read as evidence of absence. The correct statement was always the narrower one: *no reproduction that pools before the refine convs reaches it*.
- **"Song is a contaminating domain; dropping it will raise accuracy."** Backwards. Song is the *easier* subset by 16–24 points (§2a). Dropping it would delete the easy half and lower the score. The arm was built and then cancelled before it consumed a GPU slot, on the strength of the saved predictions.
- **"The refine head is a bug."** It is a deliberate, documented deviation (cell 21's comment says so) — but it *does* mean the notebook doesn't implement the paper's Fig. 4. Reclassified from bug to unimplemented specification.
- **"The target is 94.29%."** That is the notebook's score. The paper's is 96.06%.

6. Known limitations

- **The metric is the paper's:** max-over-epochs accuracy on the very fold being reported. Optimistic by construction. Kept deliberately — changing it would make comparison to Table VII meaningless. You cannot claim to beat a number you measured differently.
- **Out-of-fold features use inner 2-fold**, not full nested CV — the affordable approximation. Training features come from models trained on half the training actors; test features from models retrained on all of them (standard stacking, as `sklearn`'s `StackingClassifier` does).
- **The paper does not state its epoch counts.** The notebook uses 15 for video, and its own fold-1 log shows validation accuracy still rising at epoch 15 (90.02 → 91.24). Video is plausibly under-trained, and the paper's max-over-epochs metric rewards longer training — so a fair comparison would need an epoch count the paper never published.
- **The actor-independent numbers in §1 are a lower bound:** they were produced with the Kinetics regression in place. The per-clip re-run will raise them.